

From: 陸行之 Andrew Lu on global semis and techs <andrewhclu@substack.com>
Subject: Micron turning from margin to capacity expansion mode, like Nvidia - beat and rest
Date: 8 hours ago

Micron turning from margin to capacity expansion mode, like Nvidia - beat and rest

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Micron turning from margin to capacity expansion mode, like Nvidia - beat and rest

就像過去一年，不管 Nvidia 財務數字多美好，短期股價也不見激情，我們覺得美光將步入 Nvidia 後塵

ANDREW LU ON GLOBAL SEMI/TECHS

SEP 30 • PAID



Executive summary

For a near term, we see the strong data center AI GPU/TPU/ASIC/CPU capex and deployment to support HBM, DDR5, NAND flash demand. However, starting 4Q26 and beyond, we have started to see the pricing hike was/is decelerating and gross margin is topping but capex growth is accelerating. Despite months of inventory MOI and capital density are still at a healthy level, we believe Micron is turning itself from gross margin expansion to capacity expansion mode. Like Nvidia in the past 12 months, no matter how good the financial numbers are ahead of analysts' estimates, Micron might continue to beat the market on numbers but rest (consolidate up/down) on share price performance in line with the change of Sox index.

短期來看，數據算力中心領域對AI GPU、TPU、ASIC及CPU的強勁資本支出與部署，仍有力地支撐著HBM、DDR5及NAND快閃記憶體的需求。然而，展望2026年第四季及以後，我們觀察到價格上漲勢頭正在放緩，毛利率已觸及峰值，而資本支出增速卻在加快。儘管庫存水位（以季度庫存量/MOI衡量）及資本密度指標（資本開支除以營收）依然健康，但我們認為美光（Micron）正從「毛利率擴張」模式轉向「產能擴張」模式。正如過去12個月的英偉達（Nvidia）那樣，即便美光的財務業績持續超出分析師預期，其股價表現也可能出現盤整（或隨SOX指數波動），不再單純因業績超預期而大幅上漲。



Key highlights

- **Pricing hike continues but was/is decelerating:** Micron reports 3Q26 sales of US\$54.23bn, up 31% qoq and 379% yoy, which was 7% above Bloomberg consensus estimates of 24% qoq. Micron guides 4Q26 sales of US\$60-63bn, up 11%-16% qoq (13% qoq as mid point) and 351% yoy, which is 9% above Bloomberg consensus estimates. *We believe the 3Q26 sales beat was largely driven by stronger than expected NAND price hike of 30% qoq (20% qoq by Bloomberg estimates) and 394% yoy and also 10% bit output qoq growth for NAND (5% qoq by Bloomberg estimates);*

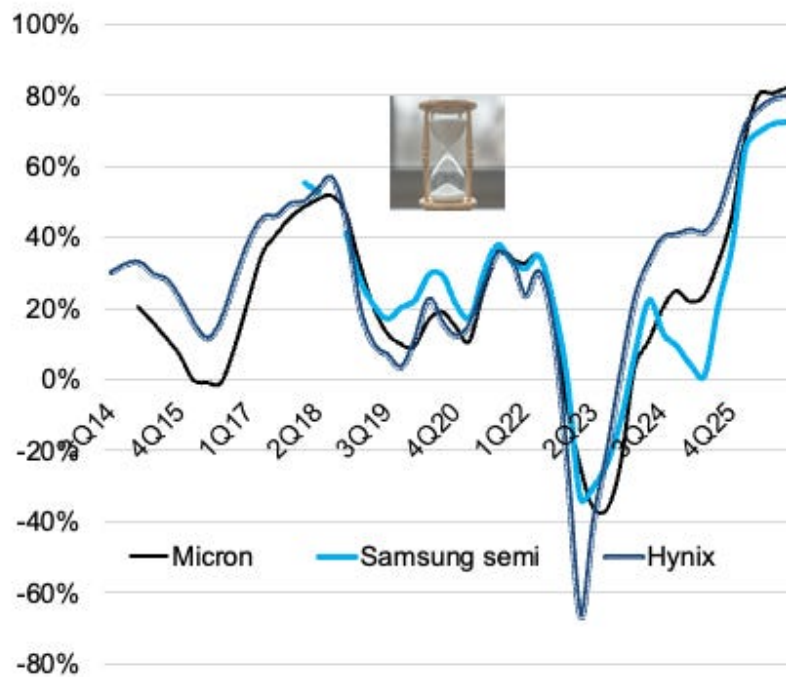
Metric	Actuals	Consensus
Analyst		
Analyst Recommendation		
Model Download		
Table Data Date	09/30/2026	09/29/2026
Highlights		
Adjusted Diluted EPS	33.42	31.83 36 4.98%
Revenue	54,229.00	51,492.31 39 5.31%
▼ Segment Revenue		
DRAM	39,771.00	38,221.69 28 4.05%
NAND	14,102.00	12,291.95 28 14.73%
Other	356.00	157.94 25 125.40%
Adjusted Gross Margin (%)	87.00	86.22 32 0.90%
▼ Adjusted R&D Expense	1,773.00	1,314.27 20 34.90%
As % of Revenue		2.59 14
▼ Adjusted Operating Income	44,636.00	42,754.63 35 4.40%
Adjusted Operating Margin (%)		82.80 28
Adjusted Net Income Margin (%)		71.20 18
▼ Capital Expenditures	-10,770.00	-10,001.15 20 -7.69%
As % of Revenue		18.55 8
▼ Free Cash Flow	33,199.00	23,526.37 16 41.11%
As % of Revenue		48.75 5
Adjusted Free Cash Flow	33,199.00	24,587.78 8 35.02%

Source: Bloomberg estimates

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- **Cyclical gross margin expansion seems topping at 85-87%:** Despite DRAM bit price hike of 17-19% qoq and NAND bit

price hike of 30% qoq, Micron reports 3Q26 gross margin of 86.8%, up only 2ppts from 2Q26's 84.6% but guides 4Q26 gross margin contraction to 85.95%, which is lower than Bloomberg's 86.7% estimates. 3Q26 operating of 80.7% and 4Q26 operating guidance of 82% are all slightly lower than Bloomberg estimates of 82-83%. Despite a price hike of 20-25% qoq for 3Q26 and likely another 5-10% qoq for 4Q26, we attributed a limited gross margin expansion for Micron to lower cost inventory is gone but higher cost of compensation new products shipment and new production are kicking in.



Source: The companies, Andrew Lu on global semis/techs

Millions of USD Months Ending	2026 Q3 05/28/2026	2026 Q4 09/03/2026	2027 Q1 Est 12/03/2026	2027 Q2 Est 03/04/2027	2027 Q3 Est 06/03/2027	2027 Q4 Est 09/02/2027	2028 Q1 Est 12/02/2027	2028 Q2 Est 03/02/2028
Income Statement (Adjusted)								
Total Revenue	41,456.00	54,229.00	56,772.90 39	62,104.21 39	66,443.49 39	70,608.44 36	72,839.11 27	73,160.26 23
Cost of Revenue	6,257.00	7,025.00	7,581.84 27	8,025.66 27	8,664.43 27	9,263.28 27	10,322.17 20	10,965.10 17
Gross Profit	35,199.00	47,204.00	50,652.62 29	54,316.78 29	58,264.04 29	61,205.25 29	62,128.63 21	61,027.46 18
Gross Margin (%)	84.90	87.00	86.73 31	86.88 30	86.75 31	86.48 30	86.25 21	85.37 19
Total Operating Expenses	1,518.00	2,568.00	1,828.62 24	1,895.54 24	1,972.90 24	2,139.70 24	2,436.72 17	2,347.37 15
Research & Development	1,186.00	1,773.00	1,449.20 21	1,493.56 21	1,545.39 21	1,713.21 21	1,860.61 17	1,730.89 14
As % of Revenue	2.86		2.56 15	2.47 15	2.39 15	2.56 15	2.71 13	2.50 10
Selling, General & Administrative	338.00	792.00	400.31 18	421.90 18	438.16 18	491.71 18	535.62 14	598.38 11
As % of Revenue	0.82		0.73 14	0.72 14	0.71 14	0.75 14	0.78 13	0.88 10
Operating Income	33,681.00	44,636.00	48,559.50 36	52,053.00 35	55,634.00 35	59,146.78 32	61,106.55 22	60,536.21 19
Operating Margin (%)	81.25		83.45 28	83.68 28	83.66 28	83.36 28	82.29 20	81.03 17
Depreciation & Amortization	2,364.00	2,641.00	2,940.98 24	3,203.88 24	3,444.44 24	3,706.61 24	4,084.84 19	4,571.97 16
EBITDA	36,045.00	47,277.00	51,090.26 19	55,751.68 19	59,897.16 19	64,423.13 16	65,924.36 14	65,800.42 12
Pre-Tax Income	33,212.00	44,238.00	48,757.35 31	52,417.87 31	56,097.65 31	59,357.37 30	60,797.62 21	59,800.17 18
Income Tax Expense	5,053.00	6,781.00	7,594.18 25	8,170.22 25	8,748.20 25	9,262.18 25	9,313.39 18	9,138.89 16
Effective Tax Rate (%)	15.00		15.38 25	15.38 25	15.37 25	15.38 25	15.52 17	15.59 15
Equity in Net Income (Loss) of ...	-9.00	-46.00	-4.37 17	-4.32 17	-4.46 17	-4.54 16	-5.49 9	-5.18 7
Net Income	28,857.00	38,398.00	41,145.91 33	44,444.76 33	47,607.33 33	50,709.68 31	51,604.70 23	50,636.95 20
Net Margin (%)	69.61		71.11 18	71.38 18	71.28 18	71.09 18	70.57 15	69.09 13
Diluted Weighted Avg. Shares	1,149.00	1,149.00	1,143.41 27	1,139.86 27	1,136.18 27	1,132.43 27	1,130.14 19	1,121.97 15
Diluted EPS	25.11	33.42	36.02 38	38.94 38	41.81 38	44.39 35	46.73 23	46.27 20

Source: Bloomberg estimates

- **Data center HBM and NAND (KV cache offload) customers were the key driver:** Despite 3Q26 core data center business unit sales only accounted for 33% of sales (28% of sales in 2Q26), the business sales increased 56% qoq in 3Q26 to US\$18bn, which was 59% above Bloomberg estimates. By combining cloud memory business and core data center business, Micron reports total data center sales of US\$34.29bn, up 36% qoq and 460% yoy by accounting global data center semiconductor sales of 12.6%, up 1ppts from 2Q26's 11.7% and up 6ppts from 3Q25's 6.6%;

Quarterly Business Unit Financial Results

	FQ4-26	FQ3-26	FQ4-25
Cloud Memory Business Unit			
Revenue	\$ 16,283	\$13,769	\$4,543
Gross margin	83%	83%	59%
Operating margin	76%	78%	48%
Core Data Center Business Unit			
Revenue	\$18,002	\$ 11,524	\$ 1,577
Gross margin	90%	87%	41%
Operating margin	85%	83%	25%
Mobile and Client Business Unit			
Revenue	\$ 13,114	\$ 11,521	\$3,760
Gross margin	90%	87%	36%
Operating margin	88%	86%	29%
Automotive and Embedded Business Unit			
Revenue	\$ 6,824	\$ 4,634	\$1,434
Gross margin	84%	79%	31%
Operating margin	79%	75%	20%

Source: Micron

- **Healthy months of inventory:** Micron reports 4.33 months of inventory, up 8% and 8% yoy;
- **Healthy capital density:** Micron raised 3Q26 capex 42% qoq and 96% yoy to US\$11.1bn to drive up the capital density to 16% in 3Q26 from 2Q26's 13% but still much lower than 3Q25's 30%. This suggests the company's capex boost still remaining at a healthy level;
- **Expanding Strategic Customer Agreements SCA:** In addition, Micron signed 26 strategic customer agreements

(raised from 16 in 2Q26) covering 35% of sales throughout 2030 (vs. 20% of DRAM and 33% of NAND volume in 2Q26), with floor prices ensuring robust gross margins;

- **Risks for us to monitor:** Till now, we believe Micron and other major memory vendors are managing capex, supply, and inventory well below demand extremely well. However, we will be watching closely on how much inventories had been piled up at the module makers and customers due to concerning pricing hike. In addition, Micron and other memory makers have been pushing its DRAM/NAND price to a painful level for customers and pushing its gross margin at historical high level of 87% for 3Q26. We thus see limited upside on ASP to boost sales and limited upside on gross margin to boost EPS for the next few quarters. In addition, by raising capex to boost bit growth and sales growth might result in depreciation cost increase to affect the gross margin improvement.

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From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: WEEKLY WEBCAST: On Inflation, Capital Spending & Economic Resilience
Date: 18 hours ago

WEEKLY WEBCAST: On Inflation, Capital Spending & Economic Resilience



YARDENI QUICKTAKES

WEEKLY WEBCAST: On Inflation, Capital Spending & Economic Resilience

By Ed Yardeni • 30 Sept 2026

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Today, Ed and Elias take a deep dive into core inflation, i.e., minus volatile food and energy prices. The Fed's preferred measure, the core PCED, justifies tightening monetary policy—both the recent September rate hike and presumably future ones provided that underlying inflation remains elevated. ... The alternative measure, the core CPI, is structured differently, causing it to diverge from the core PCED in response to price changes in AI-related spending, financial services, and shelter. ... The US

economy should remain resilient during this tightening cycle, as GDP growth has become increasingly desensitized to interest rates. That's one reason the recession widely expected in 2022 and 2023 never happened.



Exclusive Early Access for Paid Members: Below, you'll find Dr. Ed's latest webcast. Paid members can enjoy immediate access to the video. This content will become available to the public at a later date—don't miss out on early insights, consider upgrading today!



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From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: Prospect of Accelerating Wage Growth In A Strong Economy
Date: 5 hours ago

Prospect of Accelerating Wage Growth In A Strong Economy



YARDENI QUICKTAKES

Prospect of Accelerating Wage Growth In A Strong Economy

By Ed Yardeni & Elias Griepentrog • 29 Sept 2026

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I. Is Wage Growth Accelerating?

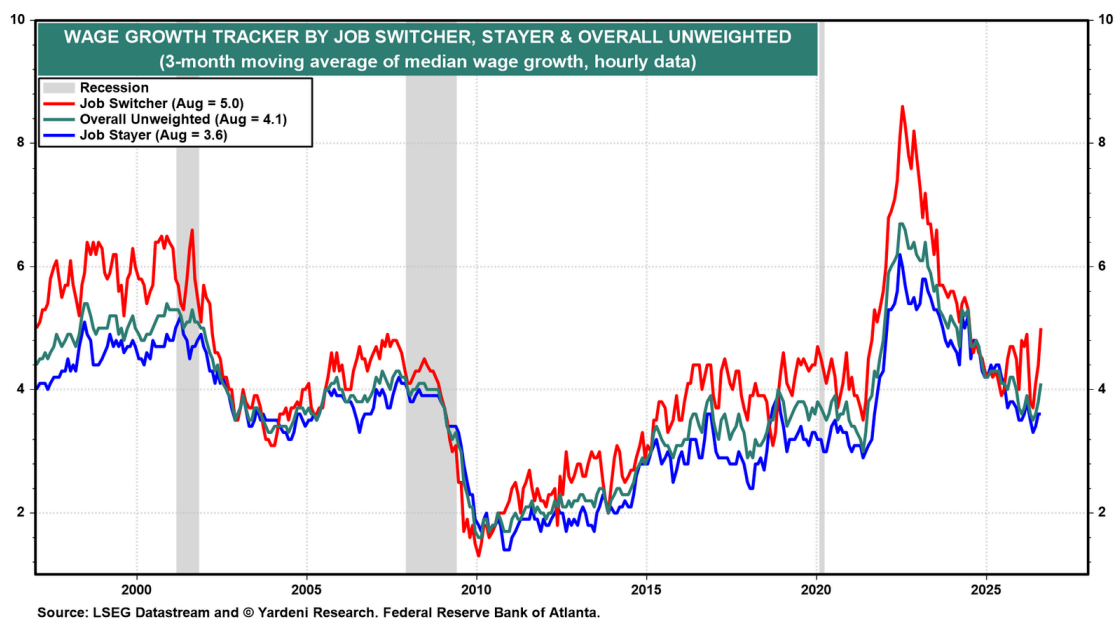
Wage growth signals labor market conditions. Just as shortages of goods tend to push prices higher, worker shortages tend to push wages higher. When labor demand outstrips labor supply, employers must compete more aggressively for workers, resulting in faster wage growth.

Former Fed Chair Jerome Powell's framework for assessing labor-market tightness compares labor demand, using the sum of household

employment plus job openings, with the labor force (chart). By that measure, demand has slightly exceeded supply for four straight months, the longest such streak since March 2023. Powell's measure of excessive labor demand correlates positively with the Atlanta Fed's Wage Growth Tracker, which may be starting to increase (chart).

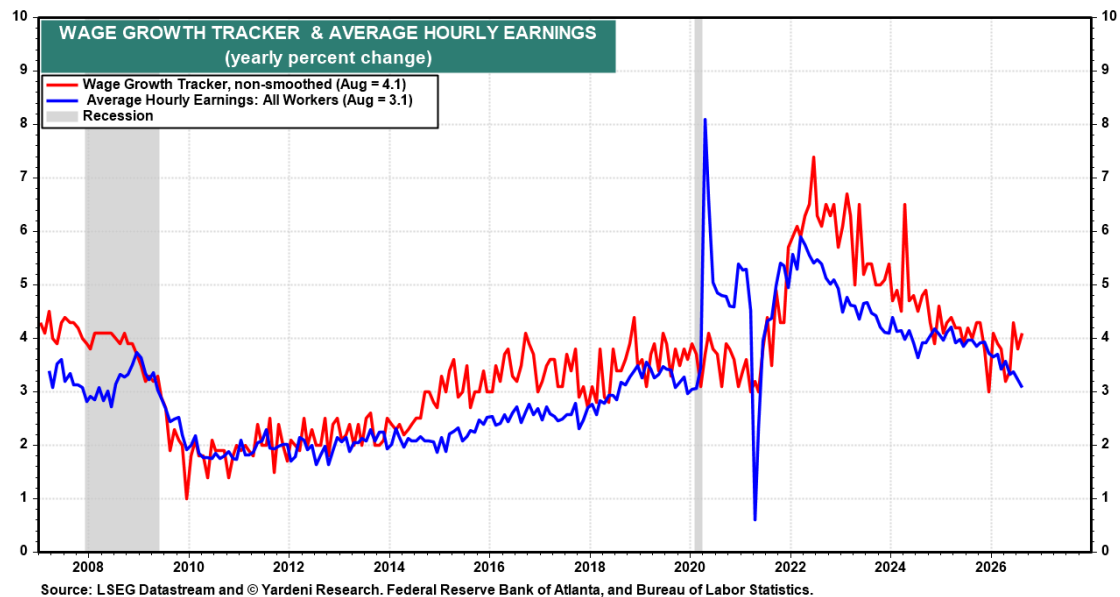


The Wage Growth Tracker rose to 4.1% y/y in August, the fastest pace since September 2025 (chart). Wage growth for job switchers accelerated to 5.0%, the strongest since June 2024.

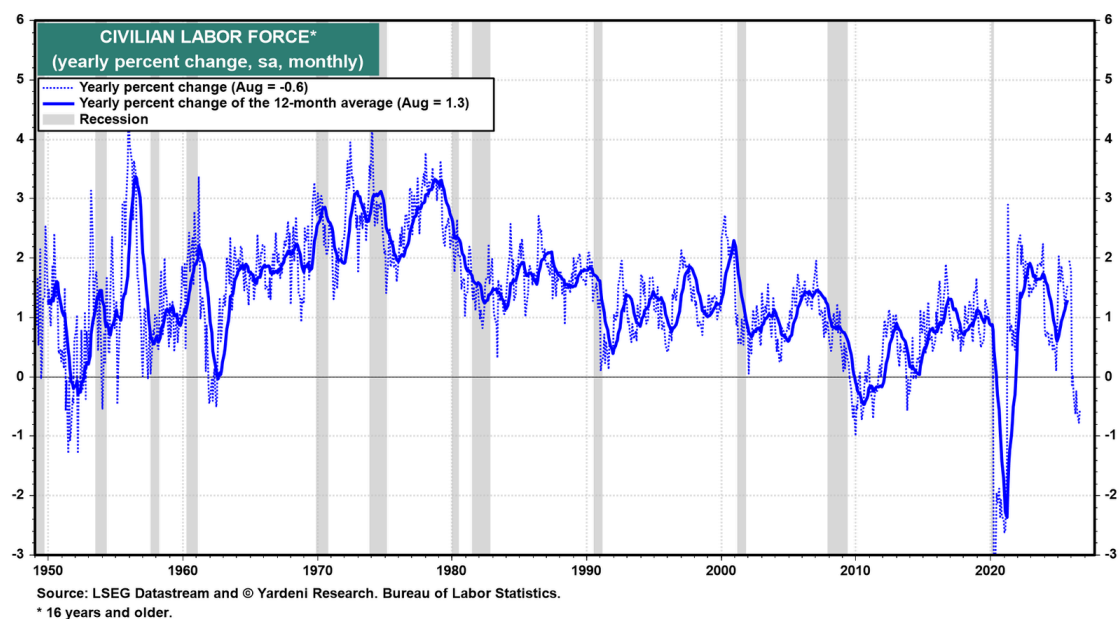


Average hourly earnings growth slowed to 3.1% y/y in August, the weakest pace since 2021 (chart). We place greater weight on the Wage Growth Tracker because it is less affected by workforce composition shifts. For example, the retirement of higher-paid Baby Boomers can

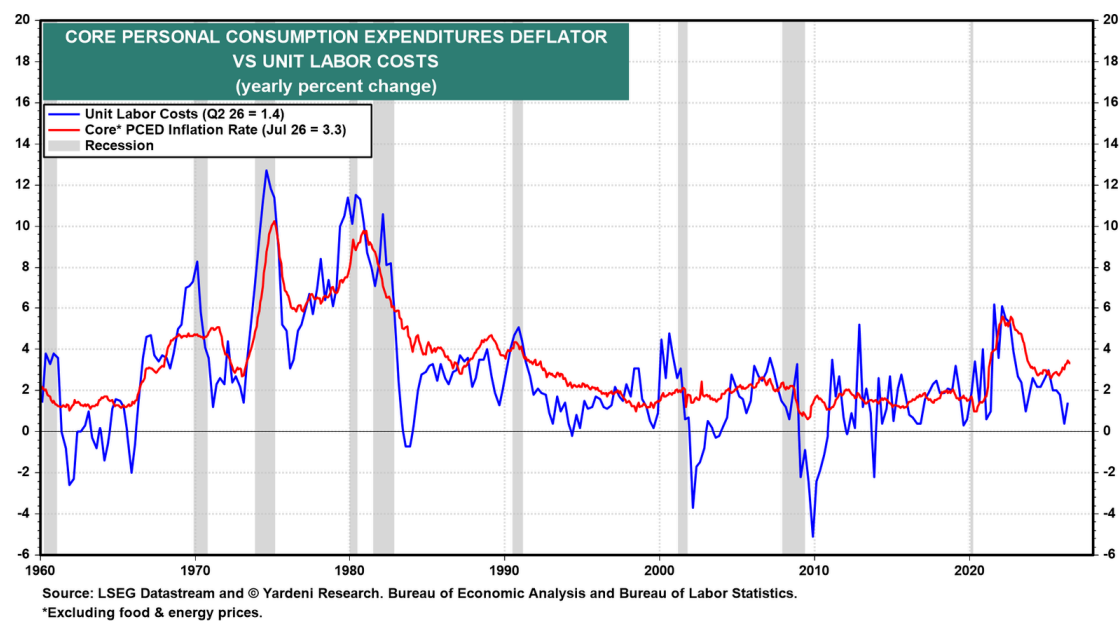
depress average hourly earnings even when underlying wage growth remains firm.



So what's the verdict? The labor market isn't tight enough yet to trigger a significant acceleration in wage growth. However, labor demand has been firming while labor supply is being constrained by tighter immigration policies and the ongoing retirement of Baby Boomers (chart). As a result, wage growth is likely to move higher as labor-market conditions continue to tighten.



A sustained pickup in wage growth would add to the list of inflationary pressures that already includes energy, tariffs, and the AI buildout. For now, the good news is that unit labor cost (ULC) inflation rose just 1.4% y/y in Q2 (chart). ULC is the ratio of hourly compensation to productivity.

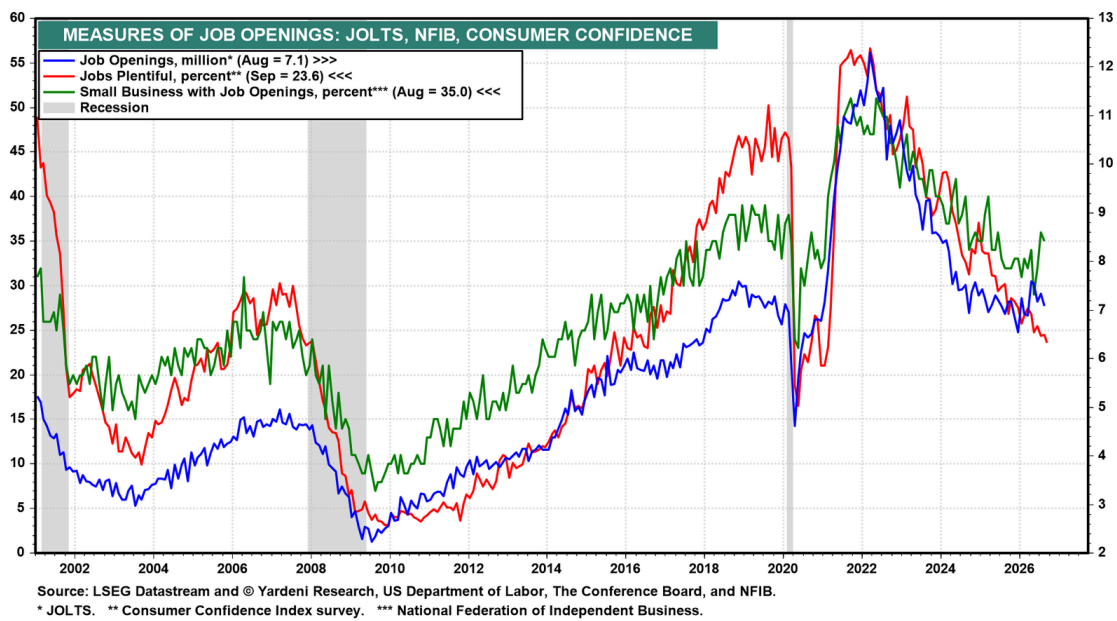


II. US Economy

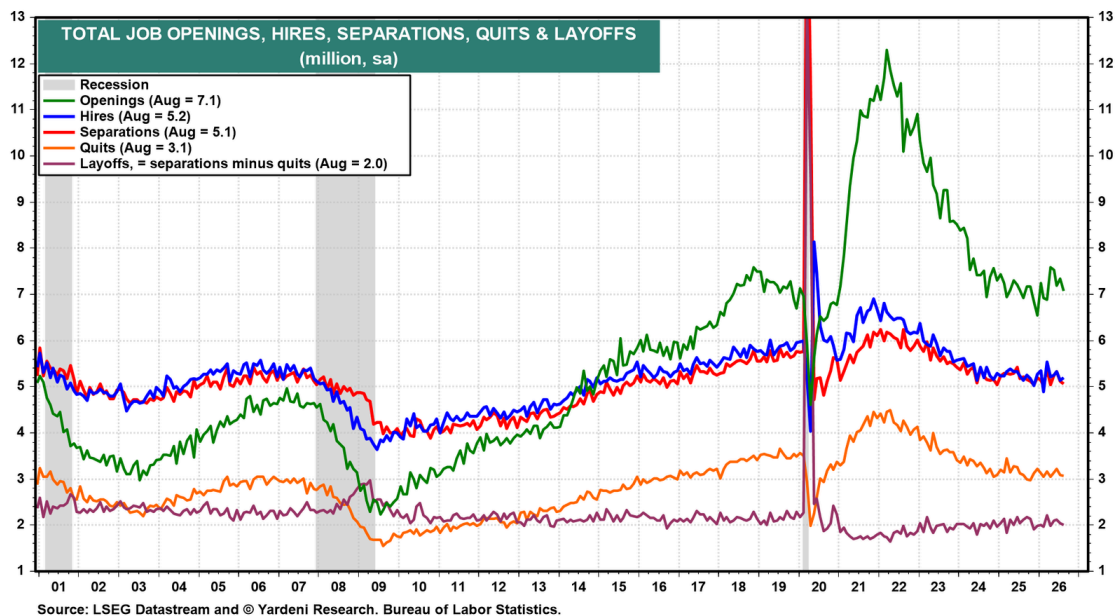
Recent economic indicators continue to confirm that the US economy is booming. That's great for corporate earnings. On the other hand, it is one reason bond yields are rising. In our view, bond yields are not currently high enough to slow the economy, which is less interest-rate sensitive than it was in the not-so-distant past.

Consider the following:

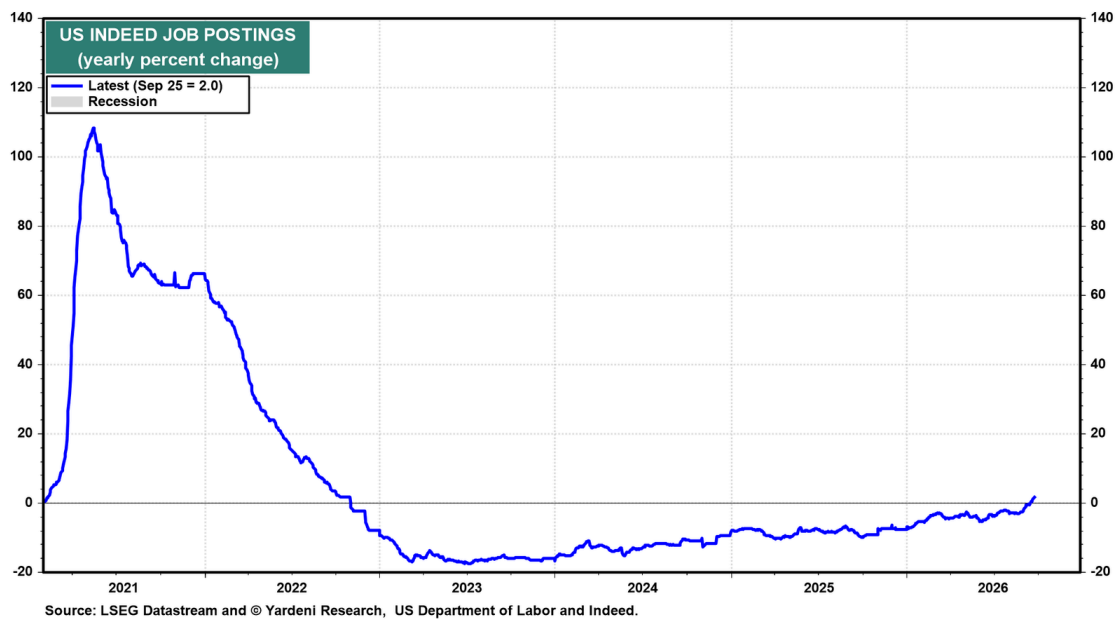
(1) JOLTS. Job openings fell to 7.1 million in August, the lowest since March (chart). We wouldn't read too much into the monthly decline, however, given the series' volatility. The six-month moving average rose to its highest level since March 2023 and has increased for five straight months. Similarly, 35% of small businesses reported unfilled positions in August, well above the historical average of 24%.



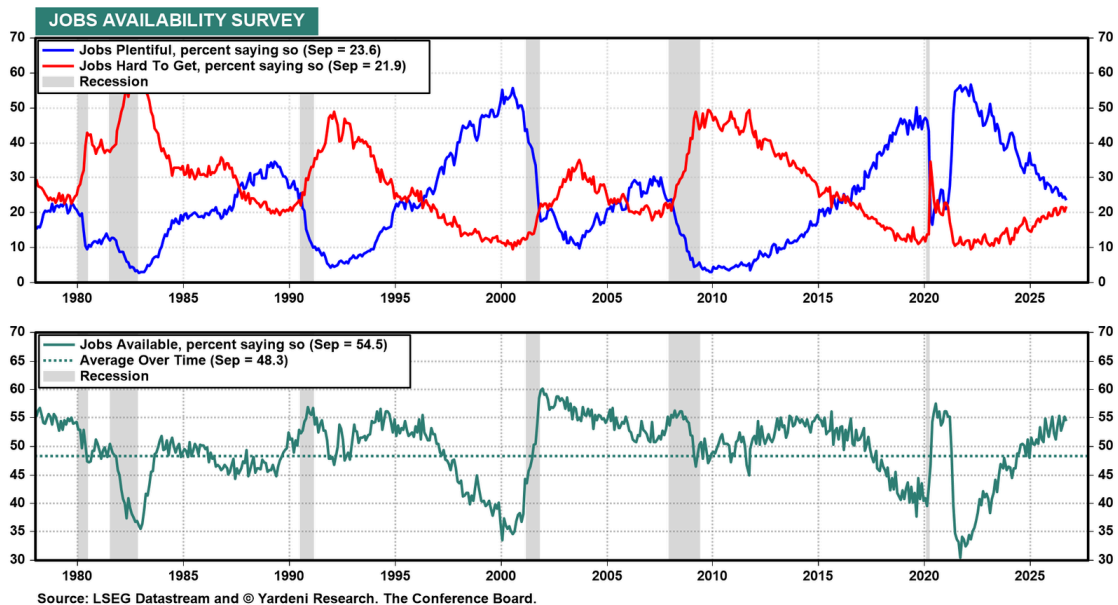
Contrary to the widely accepted "no-hires-no-fires" description of the labor market, hires totaled 5.2 million in August, slightly exceeding separations (layoffs plus quits) (chart).



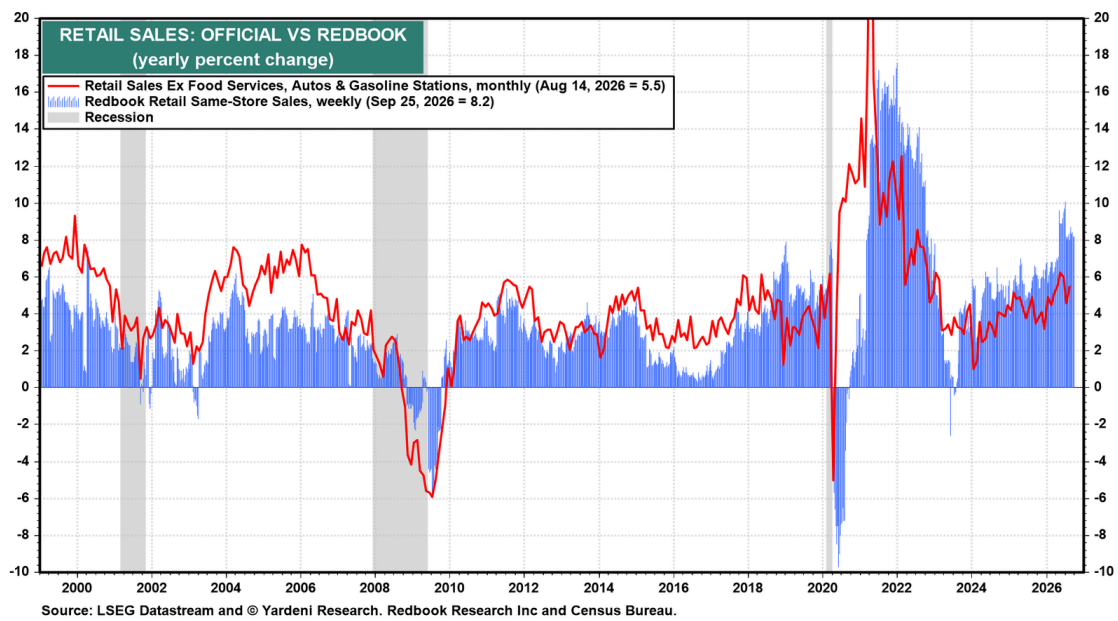
Indeed job postings turned positive on a y/y basis for the first time in four years, suggesting that labor-market activity continued to strengthen in September (chart).



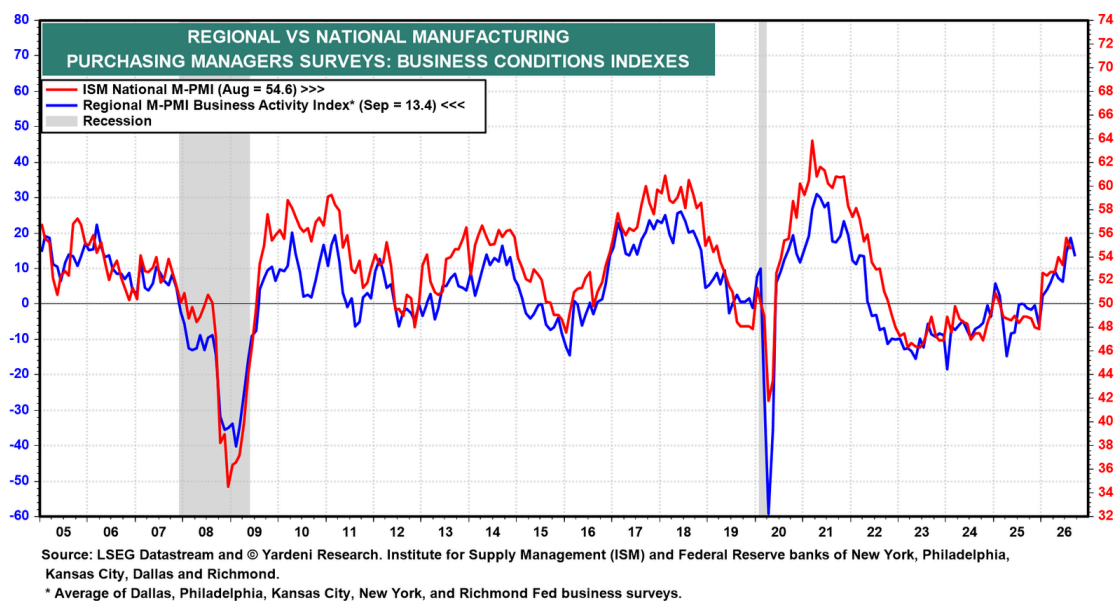
(2) Consumer confidence. How do consumers perceive the current state of the labor market? September's Conference Board data show 54.5% of respondents say jobs are available, above the long-run average of 48.3% (chart). Those saying jobs are hard to get rose from 20.3% to 21.9%.



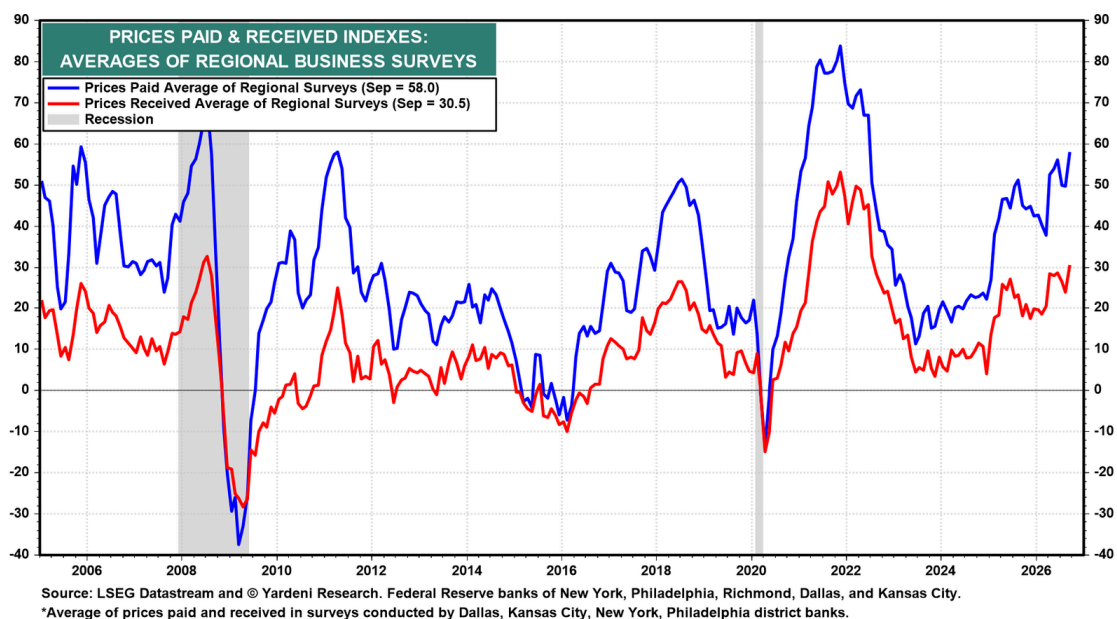
(3) Consumer spending. As expected given the ongoing strength in the labor market, consumer spending remains robust. Redbook same-store retail sales rose 8.2% y/y during the week ended September 25, remaining well above the 2025 average of 5.8% (chart).



(4) Manufacturing. Manufacturing activity remains strong too. The average of the five regional Fed manufacturing surveys remained elevated at 13.4 in September, suggesting that the national ISM M-PMI likely remained solidly in expansion territory during the month (chart).



Meanwhile, the average prices paid and received measures rose to their highest levels since 2022, indicating that upward pressure on consumer prices is likely to persist (chart).



III. Fed Speak

Not surprisingly given the latest news, two voting FOMC members recently made hawkish comments.

New York Fed President John Williams said that "if the economy evolves in a manner broadly consistent with my forecast, one further upward adjustment of the federal funds target range may be appropriate late this year." The comment suggests that Williams still expects another rate hike but is willing to be patient, potentially waiting until December before acting.

Similarly, Fed Governor Michael Barr said that "in my base case, further policy adjustments are likely to be needed to ensure inflation comes down to target in a timely fashion."

Our view remains that the Fed should at least reverse last year's three 25bps "insurance" cuts. When making those cuts, Fed policymakers underestimated the economy's strength and were overly confident that inflation was on a sustainable path back to the 2.0% y/y official target. Two additional rate hikes would better align monetary policy with the current balance of risks to the Fed's dual mandate.



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From: 陸行之 Andrew Lu on global semis and techs <andrewhclu@substack.com>
Subject: OpenAI is narrowing ARR gap with Anthropic by introducing more models at competitive prices
Date: 9 hours ago

OpenAI is narrowing ARR gap with Anthropic by introducing more models at competitive prices

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OpenAI is narrowing ARR gap with Anthropic by introducing more models at competitive prices

在不斷加碼更新有價格競爭力的模型，OpenAI 正在縮短與 Anthropic 在ARR的差距

ANDREW LU ON GLOBAL SEMI/TECHS
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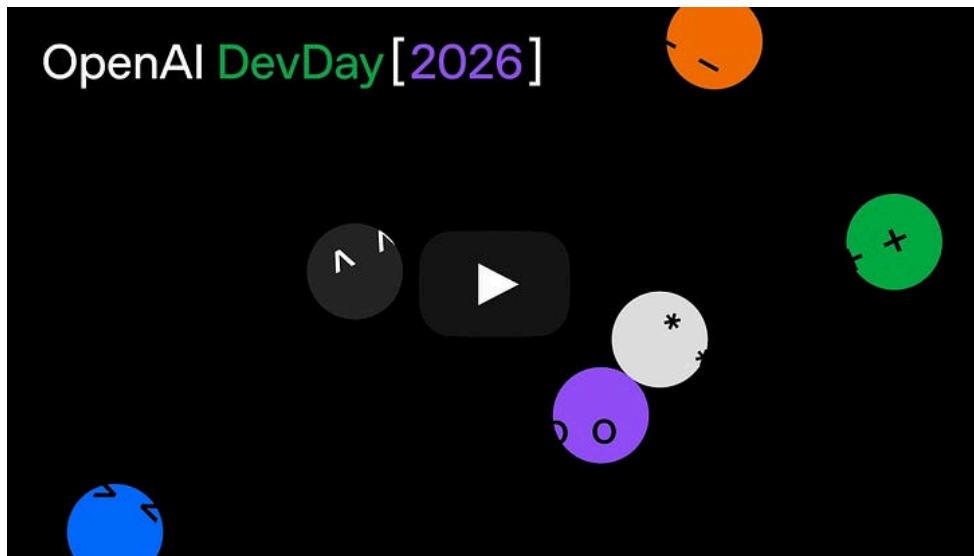


What's new?

9 月 29 日（舊金山時間），在宣佈無限期擱置旗艦模型 GPT-6.1 Astra 僅 24 小時後，OpenAI 執行長 Sam Altman 在舊金山 DevDay 2026 開發者大會上發表：

- **更新模型 GPT-6.1 Sol (vs. GPT-6.0 Sol)**: 並定位為「以五分之一的價格提供逼近 Astra 級智能的工作主力」，定價維持每百萬輸入 \$2.00、輸出 \$10.00（快取 \$0.20），配備 105 萬 Token 超長上下文與 12.8 萬 Token 最大單次輸出，全面迎戰 Anthropic 前一日發布的 Sonnet 5.5（輸入 \$2.00 / 輸出 \$10.00（快取輸入 \$0.20，批次半價））；
- **重磅推出全天候自主代理「Dots」**: Dots 是一套具備持續工作能力（Always-on）的個人 AI 代理系統，為每位用戶在雲端分配專屬虛擬機與獨立瀏覽器，能夠在背景自主執行跨軟體專案跟進、主動通話與代碼重構；同時推出團隊協同平台「ChatGPT Space」；
- **商業化變現升級**: 首度推出面向高算力專業用戶的 **\$500/月 Pro 訂閱套餐**，並將 Dots 全面整合進企業後台（因監管審查，歐洲與英國暫緩上線）。

同時根據 The Information 報導，OpenAI's ARR has jumped 70% qoq from beginning of the third quarter (US\$41.3bn in July) to US\$70bn by end of September.



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Positively affect what supply chain vendors and why?

雲端虛擬化與專用推論伺服器供應商（微軟 **Azure**、戴爾 **Dell**、美超微 **Supermicro**）：Dots 代理需要為數百萬用戶維持全天候常駐的雲端虛擬機與瀏覽器實例，使推論伺服器的DDR5 記憶體占用與雲端 CPU 負載激增數倍，又一 CPU：GPU佔比增加的支撐（請參考之前深度報告：[Party over for AI players or just a short term valuation reset even after AMD raised its AI TAM again?](#)）。

企業級端點工作流自動化與 API 整合平台：Dots 連接超過 4,000 款企業軟體，帶動專用 API 網關與企業內部系統數據清洗市場爆發。

縮短與 Anthropic 在ARR的差距：May ARR (Anthropic: US\$47bn vs. OpenAI: US\$33bn), July ARR (Anthropic: US\$65bn vs. OpenAI: US\$41.3bn), Sep ARR (Anthropic: N/A vs. OpenAI: US\$70bn)。

Units in mn	2023	2024	2025	2026	2027
General/AI CPU			28.40	36.10	52.60
General server CPU			24.60	30.70	42.20
AI server CPU			3.80	5.40	10.40
YoY in %					
x86 and ARM CPU				27%	46%
General server CPU				25%	37%
AI server CPU				42%	93%
%					
General server CPU			87%	85%	80%
AI server CPU			13%	15%	20%
x86 and ARM CPU					
x86 CPU			23.00	26.30	34.10
ARM CPU			5.40	9.80	18.50
YoY in %					
x86 CPU				14%	30%
ARM CPU				81%	89%
%					
x86 CPU			81%	73%	65%
ARM CPU			19%	27%	35%
By companies in mn units					
Intel x86 CPU			16.00	17.50	20.50
AMD x86 CPU			7.00	8.80	13.60
Nvidia Grace/Vera/Rosa			2.10	4.70	5.80
Amazon Graviton			2.10	2.30	4.20
Google Axion			0.60	1.30	4.00
Msft, ARM, others			0.60	1.50	4.50
YoY in %					
Intel x86 CPU				9%	17%
AMD x86 CPU				26%	55%
Nvidia Grace/Vera/Rosa				124%	23%
Amazon Graviton				10%	83%
Google Axion				117%	208%
Msft, ARM, others				149%	200%
%					
Intel x86 CPU			56%	48%	39%
AMD x86 CPU			25%	24%	26%
Nvidia Grace/Vera/Rosa			7%	13%	11%
Amazon Graviton			7%	6%	8%
Google Axion			2%	4%	8%
Msft, ARM, others			2%	4%	9%

Source: DigiTimes

Negatively affect what supply chain vendors and why ?

純依靠人工排程的傳統 **RPA**（機器人流程自動化）軟體商：OpenAI 直接推出具備自主環境理解能力的 Dots，將大幅蠶食傳統 RPA 軟體的續約率。

歐洲受限市場的本土 **AI 服務商**：由於 Dots 因歐盟 AI 法案與英國安全審查暫緩進入歐洲，歐洲企業在獲取次世代自主代理工具上面臨競爭代差。

AI 代理人進入紅海商品市場：在 OpenAI 併購Openclaw再推出 Dots, Gemini 推出Sparks, Anthropic 推出 Claude managed agents, xAI Grok 推出Bot, Meta 推出 Muse 之後，雲端 AI 代理人已經變成各家大模型的基本配備，未來的觀察重點是哪家公司有足夠的數據中心伺服器 CPU 數量，哪家公司能在 AI 代理人執行任務時保護企業及個人用戶的隱私及帳戶，及安全執行付款任務（企業：Microsoft，消費者：谷歌，蘋果）。

Our outside the box non-consensus thinking

市場常將 Dots 視為另一個「聊天機器人升級版」。非共識深度洞察在於這標誌著大模型計算架構從「無狀態請求（Stateless Requests）」向「有狀態數位工人（Stateful Digital Worker）」的質變。過去的模式調用是「一問一

答」，算力只在回答的那幾秒鐘被消耗；而 Dots 擁有自己的雲端虛擬機和長期記憶體，即使人類離線睡覺，代理仍在背景持續運算、監控網絡、編寫代碼。這意味著：全球 AI 推論算力的負載曲線將徹底被「熨平」，推論算力不再有白天與夜晚的高峰低谷之分，而是成為類似水電的 24 小時恆定基荷（Baseload）。這將直接推高資料中心的年化稼動率（Utilization Rate），這對數據中心算力基礎建設的有效投資報酬率（ROIC）構成結構性支撐。

Our advises to investors

從最新趨勢變化來看，AI 算力需求及投資還在持續擴大，這有利於伺服器半導體產業鏈，這新聞短期有利於 OpenAI 產業鏈，尤其是提供算力底座的微軟（MSFT）及 Oracle；逢低配置受惠於常態化推論放量的先進製程晶圓代工龍頭台積電（TSMC）；審慎看待護城河薄弱的單一功能流程自動化公司。

Risks to our analysis

Dots 在背景運行中再次觸發未經授權的跨站越權漏洞，引發美國聯邦貿易委員會（FTC）或司法部的緊急強制暫停令。

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From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: Thoughts On Global Government Debt With A Focus On The US
Date: 5 hours ago

Thoughts On Global Government Debt With A Focus On The US



YARDENI QUICKTAKES

Thoughts On Global Government Debt With A Focus On The US

By Ed Yardeni & Elias Griepentrog • 28 Sept 2026

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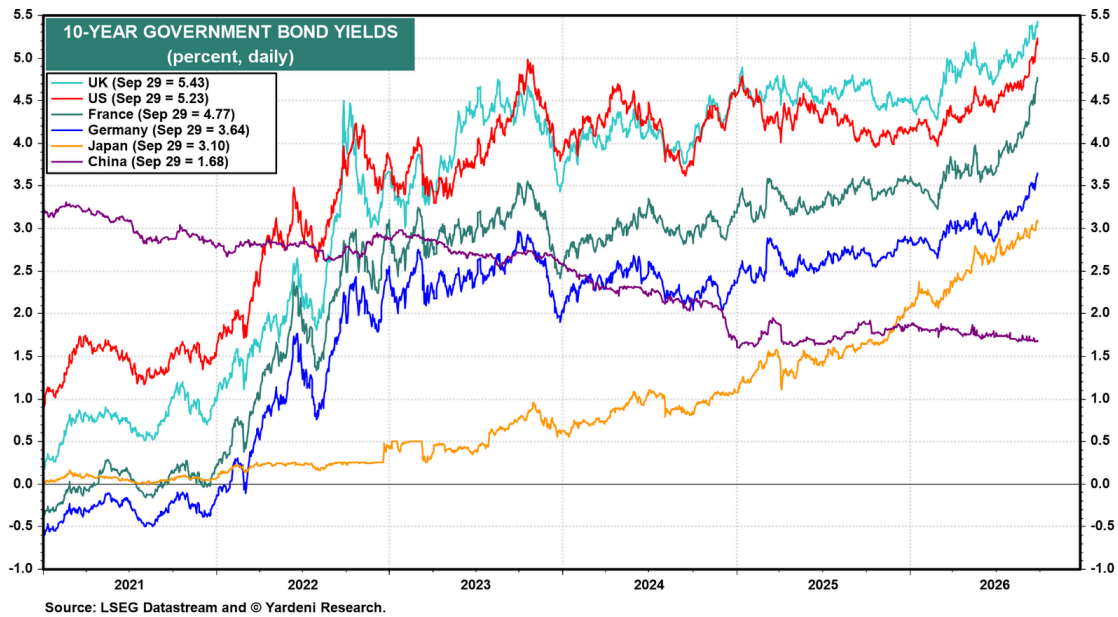


Photo by [Alice Pasqual](#) / [Unsplash](#)

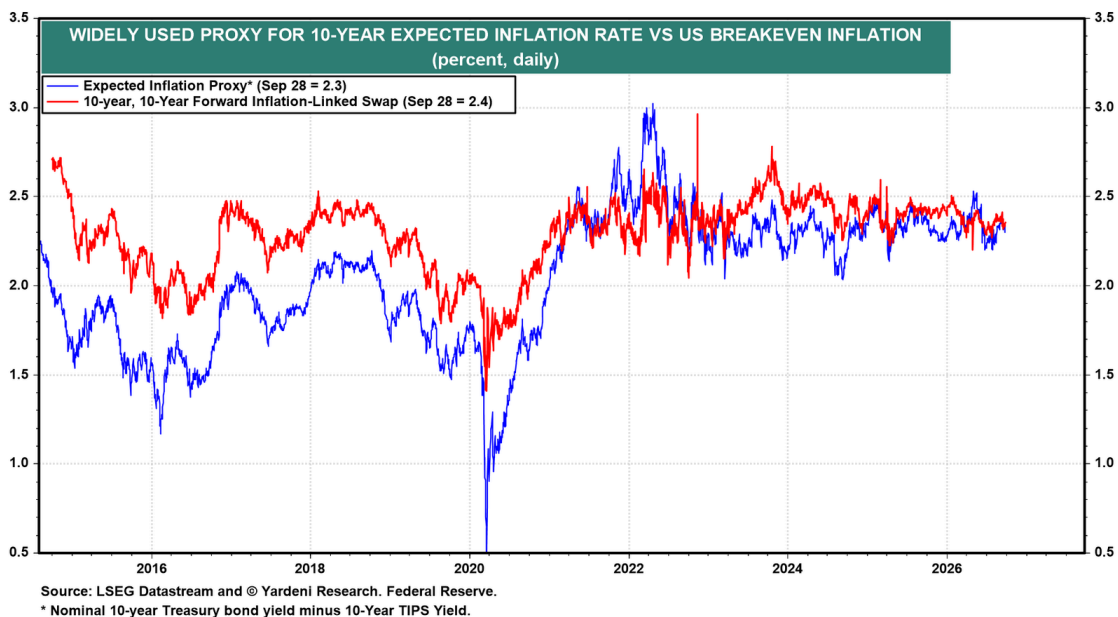
I. Revenge of the Bond Vigilantes?

We still have a 70% subjective probability for our bullish base-case Roaring 2020s scenario. The remaining 30% covers all the possible bearish scenarios. We monitor those possibilities closely with our Worry

List. Our main worry right now is the significant rise in bond yields worldwide this year (chart).

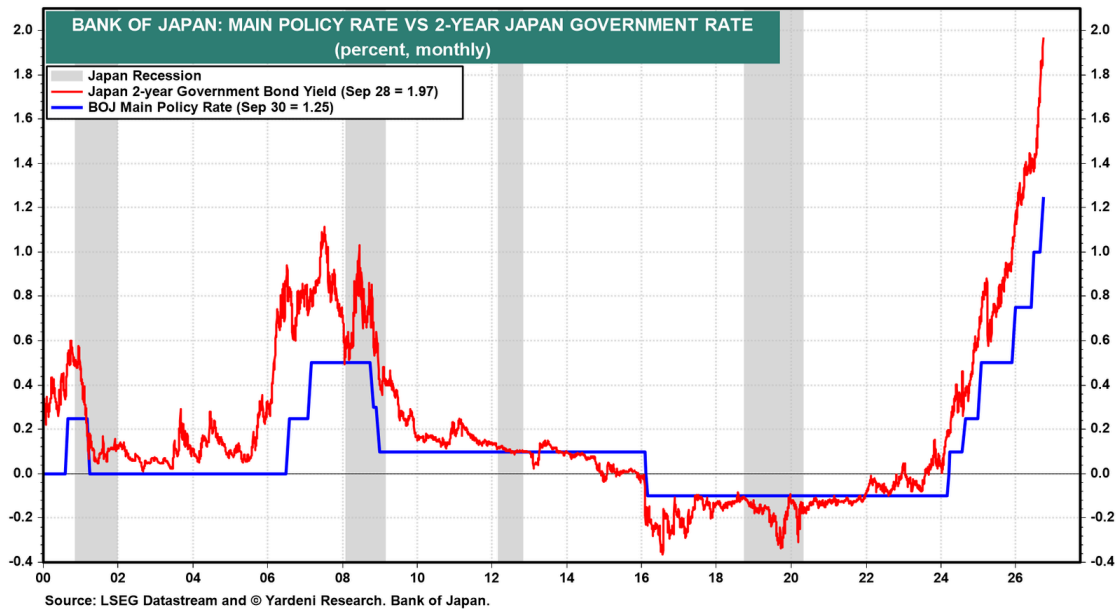


The higher global bond yields may be due to higher inflation, driven by the jump in oil prices following the Middle East war that began in late February. That's not confirmed by US breakeven inflation rates, which remain surprisingly subdued (chart)! Nevertheless, when the war ends, oil prices should drop sharply, lowering bond yields.



A more likely explanation for the global bond market rout is that the yen-carry trade is unwinding as the Bank of Japan (BOJ) raises its policy rate, forcing carry traders to sell government bonds they bought worldwide with proceeds from cheap yen loans (chart). This trade allowed many

governments run budget deficits without putting upward pressure on their bond yields. Now, the chickens have come home to roost.



Governments ran large deficits and accumulated lots of debt when the BOJ and other major central banks kept interest rates abnormally low from the Great Financial Crisis through the Great Virus Crisis. The major central banks' quantitative easing policies rigged global bond markets. The Bond Vigilantes were subdued. Now, we may be witnessing the Revenge of the Bond Vigilantes.

II. Is Bessent getting twisted?

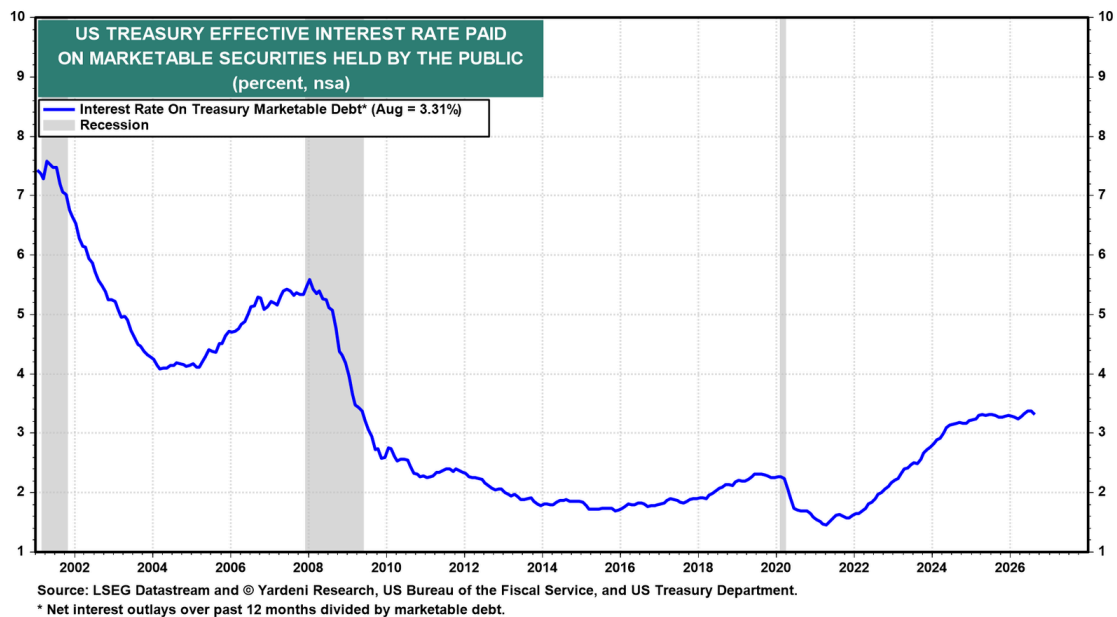
US Treasury Secretary Scott Bessent has been leaning on the BOJ to raise its official policy rate to bolster the yen, which has been very weak. He wants to make sure that Japan doesn't sell its US Treasury securities to support the yen. The problem is that a higher BOJ policy rate would probably cause the yen-carry trade to unwind faster, putting upward pressure on bond yields worldwide.

Bessent also has been gingerly implementing an "Operation Twist" in the US Treasury market by buying back bonds and issuing more T-bills to fund the purchases. If the bond market rout turns into a US debt crisis, he might have to significantly increase the size of his Operation Twist.

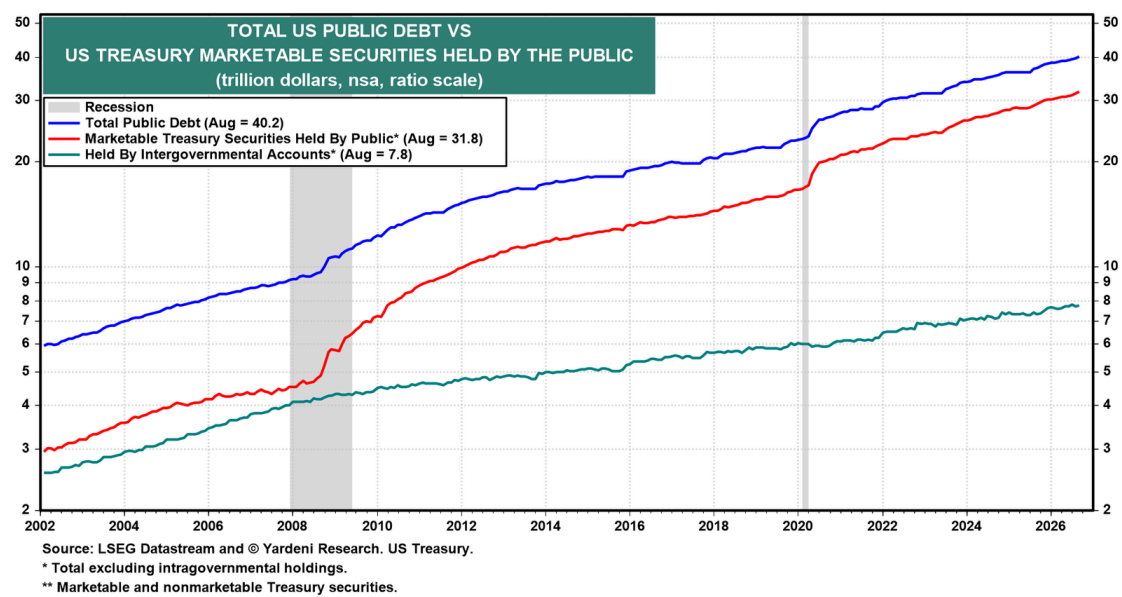
III. Is the US on an unsustainable fiscal course?

Larry Kudlow, director of the National Economic Council under President Donald Trump during his first term, invited me to speak at the White House Economic Advisers' lunch on December 12, 2018. Joining us was Jason Trennert, chairman, CEO, and chief investment strategist of Strategas. Several of the President's top economic advisers attended. I asked then Treasury Secretary Steven Mnuchin why the administration wasn't refunding the entire government debt. Mnuchin said, "We are looking into that." Nothing changed.

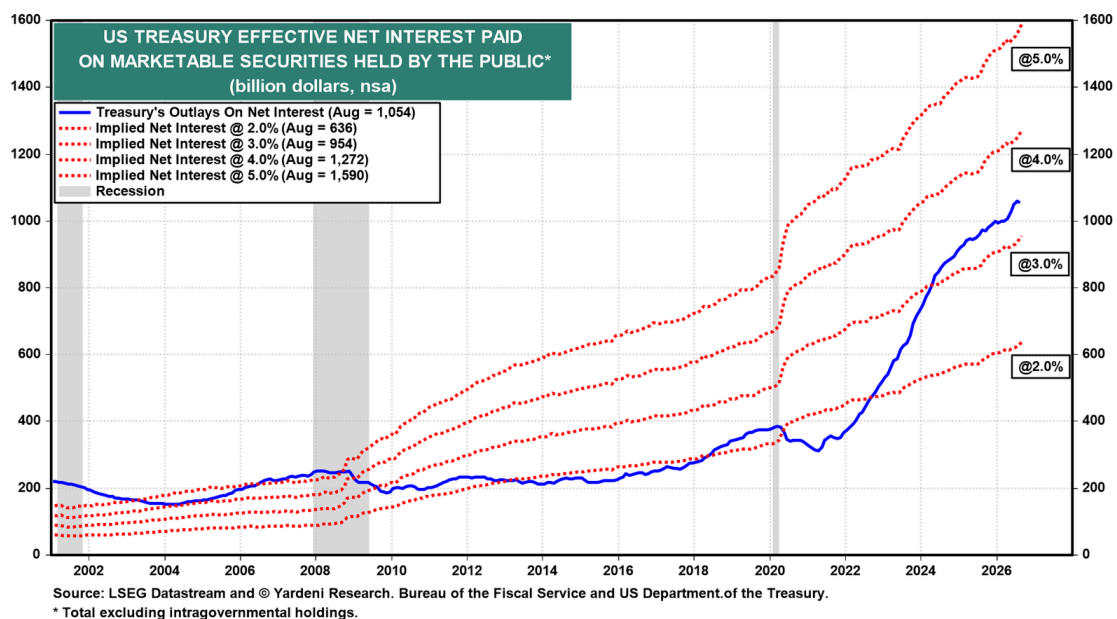
At the time, the three-month Treasury bill rate was 1.32%, and the 10-year bond yield was 2.40%. The average effective interest rate paid on the Treasury's marketable securities was just below 2.00% (chart). Now the T-bill rate is at 4.08%, and the bond yield is at 5.17%. The effective rate was 3.31% in August.



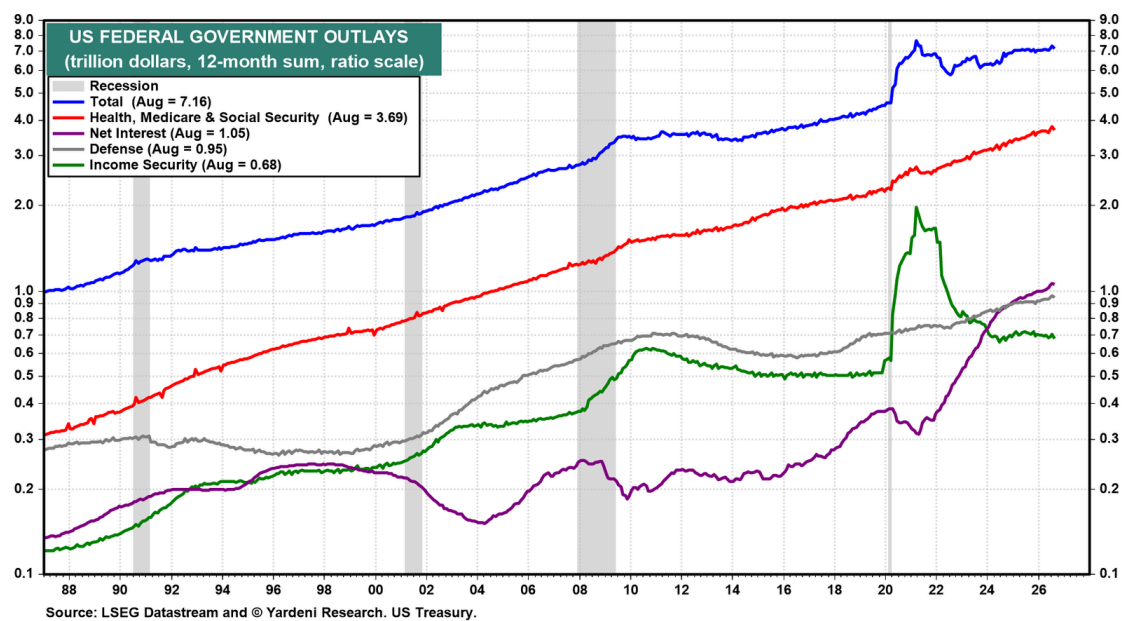
Since that lunch meeting in the basement of the White House in December 2018, federal marketable Treasury debt has more than doubled, rising from \$14.4 trillion then to a record \$31.8 trillion in August of this year (chart).



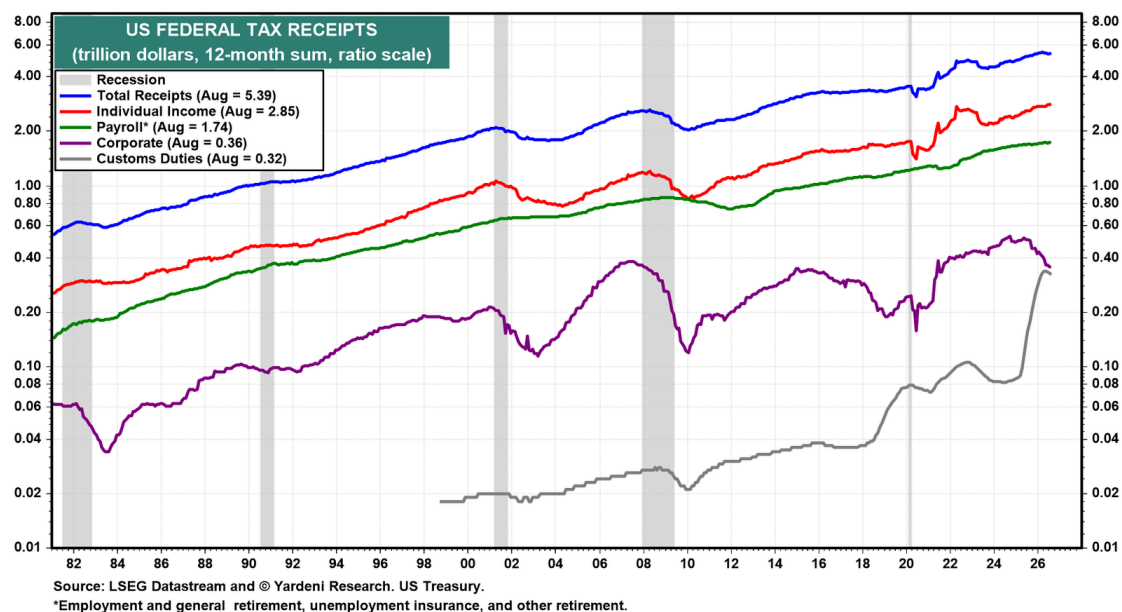
Net interest outlays on public debt held by the public soared from \$300 billion at the end of 2018 to \$1.1 trillion in August 2026 (chart). Persistently large government deficits and now higher interest rates suggest that net interest outlays are heading higher.



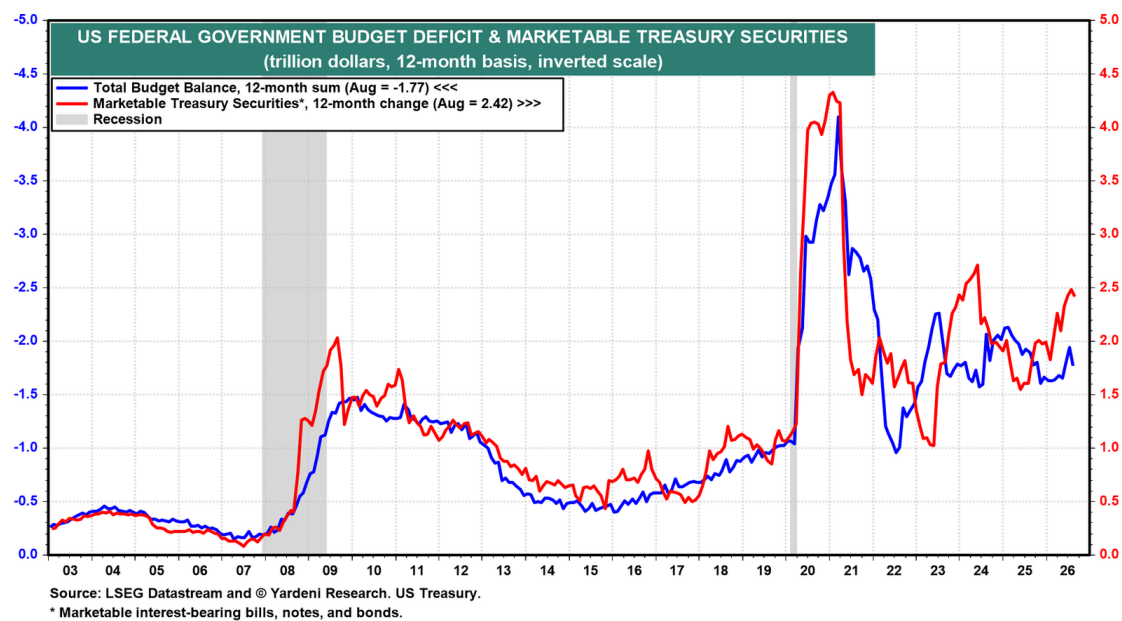
Net interest outlays now exceed both defense spending and income security outlays (chart). Towering above them all, of course, is the relentlessly rising spending on health, Medicare, and Social Security.



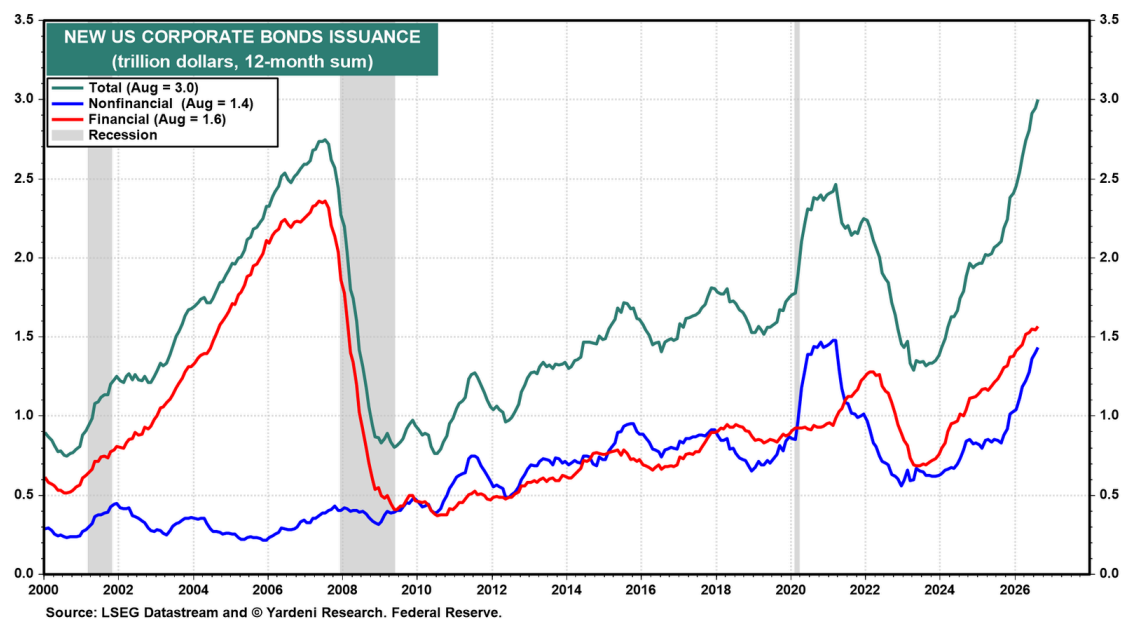
Federal tax receipts are driven mostly by individual income and payroll tax receipts (chart). Both have been increasing along with employment. Corporate tax receipts have been falling this year, probably because last year's tax bill allowed 100% depreciation.



The federal budget deficit totaled \$1.77 trillion over the 12 months through August (chart). Over that same period, net marketable Treasury securities rose \$2.42 trillion.



Meanwhile, the pace of new US corporate bond issuance has doubled since the start of 2024 to \$3.0 trillion over the past 12 months through August (chart). Interestingly, not all of it is AI-related, since financial corporations raised \$1.6 trillion.



Yes, the US government is on an unsustainable fiscal course. But Bessent believes that our Roaring 2020s scenario will save the day: Productivity-led growth should boost federal receipts, moderate inflation, and lower interest rates. Fed Chair Kevin Warsh believes so too. The Bond Vigilantes aren't cooperating. That's admittedly worrisome. Nevertheless, we expect the Roaring 2020s to prevail.



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From: 陸行之 Andrew Lu on global semis and techs <andrewhclu@substack.com>
Subject: Understanding the balance film industry a for larger AI chip (Part I)
Date: 6 hours ago

Understanding the balance film industry a for larger AI chip (Part I)

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Understanding the balance film industry a for larger AI chip (Part I)

應該是時候了解AI大晶片抗翹曲應力平衡膜產業鏈

ANDREW LU ON GLOBAL SEMI/TECHS
SEP 28 • PAID



Executive summary

雖然 Balance Film 產業還在初始冒芽階段，但當AI GPU，TPU，ASIC 越來越大顆，ABF載板及RDL 處理層數越來越厚，越來越多不同熱膨脹係數晶片及材料被封裝在一起，防翹曲技術及抗翹曲應力平衡膜就越來越重要。當未來10年，AI 算力數據中心的投資將受限於可用資金，可用電力，可用土地，可用晶片產能，除了發展困難度超高的衛星軌道數據中心，提供更高算力/更低耗能（單位面積算力耗能比較，增加每單位數據中心面積的GW算力）的大型AI訓練及推理晶片的騰籠換鳥原廠升級（舊機櫃換成新機櫃+ 光通訊 scale out/scale across），可能反而是一個可行的方向。

What is the balance film?

Balance Film（抗翹曲應力平衡膜）在先進封裝領域中，是一種專門用來「抵消」異質整合製程中因熱膨脹係數（CTE，Coefficient of Thermal Expansion）不匹配所產生之應力，進而抑制載板與晶圓翹曲（Warpage）的關鍵材料。

在進行 CoWoS, FOPLP（面板級扇出型封裝）或玻璃基板（Glass Substrate）等大尺寸封裝製程時，矽晶粒、矽/玻璃中介層，ABF載板、玻璃基板（Glass core substrate）、銅線路（RDL）以及環氧樹脂（EMC）的膨脹與收縮率各不相同。經過高溫製程冷卻後，這種應力失衡會導致嚴重的翹曲變形，造成後續微影對位偏移或基板破裂。Balance Film 透過貼合或塗佈於

基板特定區域，在前段製程中主動形成反向拉扯的「對沖」應力，將基板維持在平整狀態。



When the market started to use balance film?

我們需要了解，大型 AI GPU/TPU 晶片的 CoW 矽/玻璃中介層「有翹曲問題」不代表現在都一定會用獨立的 Balance Film。Balance Film 是一種額外的「反向應力補償材料」，主要在大尺寸、高層數、薄型化、RDL/PLP/先進封裝製程翹曲變嚴重時導入，而不是標準必備層。

AI 晶片類型	Balance Film 必要性	時間判斷
B200-Vera Rubin, TPU v8	低	已經開始面臨 warpage, 但仍靠對稱 build-up, 低CTE core, 材料配方，均熱片Lid
Vera Rubin Ultra	中	2H27 比重增加
TPU v9	中	2H27 比重增加
Rosa Feynman	高	2H28 比重增加
TPU v10	高	2H28 比重增加
CoPoS	高	2H29 比重增加

Source: Andrew Lu on global semis/techs

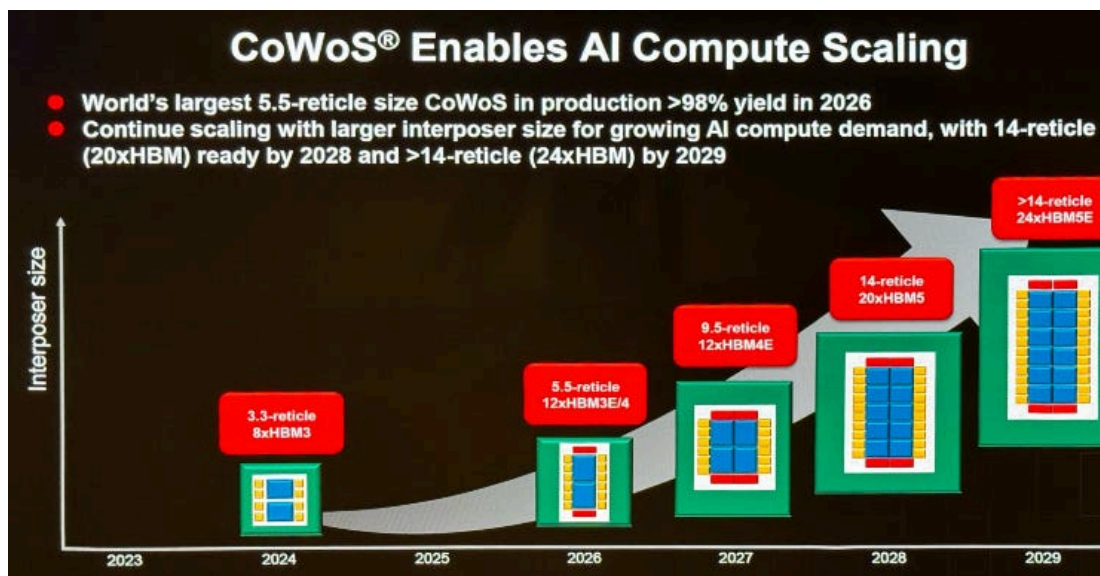
Why a larger AI chip needs it?

Chip size matters

AI GPU/TPU 的矽中介層 Silicon Interposer 將走到5.5x 光罩尺寸，晶片面積增加後，當台積電AI GPU客戶Nvidia/AMD及AI ASIC/TPU客戶亞馬遜/谷歌/Meta/微軟為了追求更高算力從2028年開始推疊邏輯電晶體及12顆HBM4E存儲記憶體，甚至20-24顆HBM5E存儲記憶體，一片12“晶圓能切出8片的5.5x reticle size (70mmx70mm=4,900mm2)晶片大小所需要的矽中介層 (Silicon Interposer)，將只能切出5片 9x reticle size (90mm x

90mm=8,100mm²) 晶片大小的矽中介層，甚至4片 14x reticle size (110mm x 110mm=12,100mm²)的矽中介層，這意思是指假設台積電客戶全面推出新一代 9x/14x reticle size 的晶片產品，即使數量不變，台積電矽中介層的產能會立刻減少40-50%，如果再考量良率因素，矽中介層產能減少50%或以上，9x/14x reticle size 的矽中介層漲價一倍是非常合理的。因為這個結構性變化，我們認為台積電對於幫助大晶片客戶採用玻璃中介層取代矽中介層，於2029年由圓轉方加速擴張CoPoS產能及量產 CoPoS 將是台積電未來10年不得不為的重大決定（請參考我們於2026年3月27日發出的深度報告：[CoWoS/FOWLP 轉 CoPoS/FOPLP 圓轉方的典範轉移及產業鏈分析](#)）。

現行大型 AI ABF substrate 已普遍需要 low-CTE core、symmetrical stack-up 等手段抑制 warpage。當矽/玻璃中介層未來超過 90mm x 90mm=8,100mm² 大小，矽晶片、矽/玻璃中介層，銅（RDL）、與環氧樹脂（EMC）之間的熱膨脹係數（CTE）差異，會導致翹曲（Warpage）呈現「非線性」的急遽惡化。當翹曲變形量超過微影設備的對焦容許度（Depth of Focus）或造成晶圓/面板破裂時，傳統的物理載具已無法壓制，就必須在製程前段導入 Balance Film。



Source: TSMC

付費訂閱戶可分享 Part I 全文，為何AI大尺寸晶片需要BF(Size, layers, mix matters)，貼在哪裡，是半導體產品材料還是製程耗材？Balance film 移除後如何防翹曲？Part II 重點

✓ Subscribed

Layers matter for ABF substrate

AI GPU/TPU I/O 數量暴增，因此 ABF 載板的 routing layer 越來越多。ABF 載板本身是 build-up dielectric，以後高階 ABF載板 可達超過 20+ layers。

Top 頂端

ABF / Cu

ABF / Cu

ABF / Cu

———— core

ABF / Cu

ABF / Cu

ABF / Cu

Bottom 底端

目前理想狀態是上下完全對稱，內部應力互相抵消，但實際 AI ABF 載板，因為不同的 power plane, routing density, copper density, interposer/RDL 只在一側，HBM/GPU/TPU 集中在頂端不可能永遠完全對稱，molding/underfill/TIM 熱介面材料（TIM，Thermal Interface Material）等材料不同，因此熱循環後會產生 bow / twist。Balance Film 的作用就是人為加入一個相反方向的 stress，使整體 stack 的 bending moment 接近零。



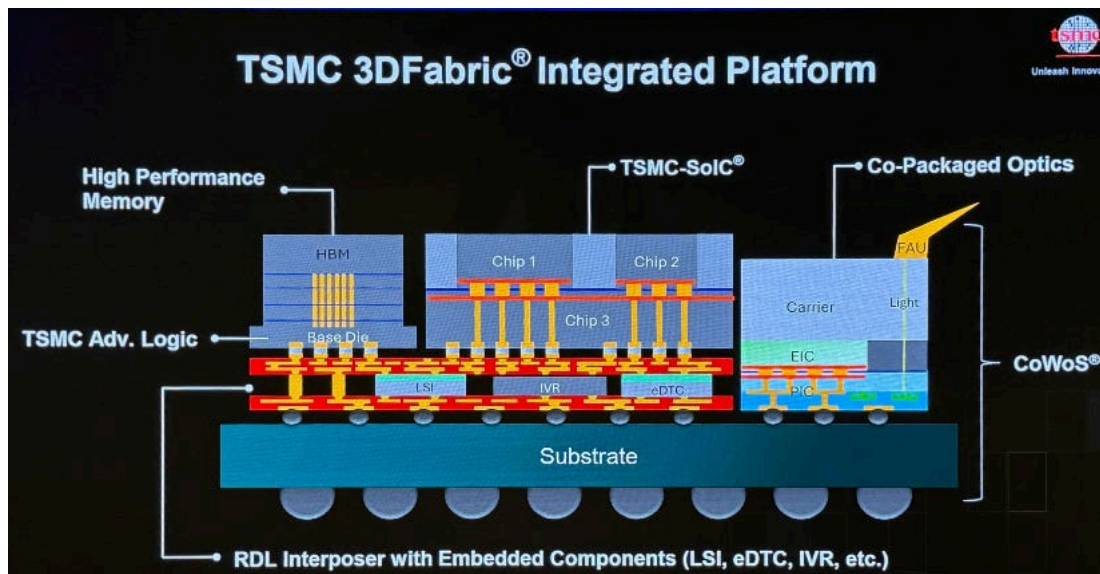
Source: TSMC

Mixed chips matter

未來的AI GPU/TPU 已經從單純 AI 加速器 + ABF substrate 變成 AI加速器 + CPO/Optics on 中介層 (EIC/PIC/FAU) + 矽/玻璃中介層 + HBM 的超級複合體，而每種材料的熱膨脹係數都不同：

- 矽：約 2.6 ppm/K
- 銅：約 16.5-17.0 ppm/K
- 玻璃：約0.5-10 ppm/k
- ABF 載板：平面方向 X-Y軸 12-16 ppm/K；垂直方向 Z軸 30-50 ppm/K

所以從室溫到 reflow 250°C 左右，應力差非常大。Ajinomoto 也把 low CTE / high Tg 列為最新 ABF 材料的重要特性，目的之一就是控制熱機械變形。



Source: TSMC

Where balance film attached to?

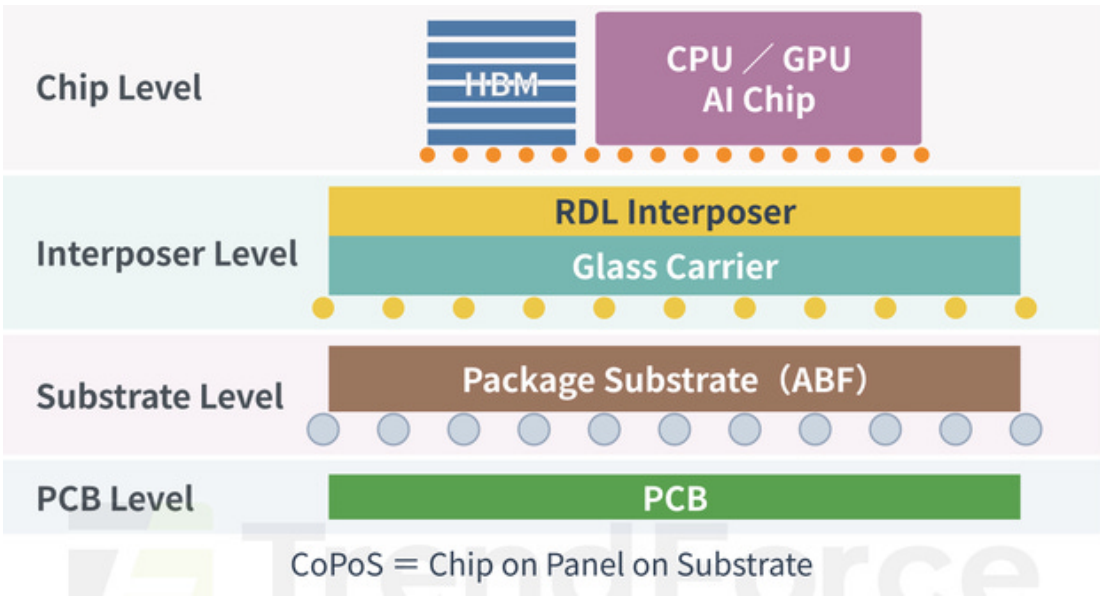
Balance Film 不是留在最終晶片產品的材料，更像是製程耗材，核心概念是「反向對沖」。因為封裝結構的正面（Front-side）堆疊了大量會產生應力的材料，所以 Balance Film 通常是貼附在矽/玻璃中介層或玻璃載體 glass carrier 的背面（未來也有機會用在 ABF 載板），並不等於CoPoS的專屬材料（請參考之前圓CoWoS轉方CoPoS 深度報告）。

根據不同的製程階段，其貼合位置與作用對象略有不同：

1. RDL（重佈線層）製程階段

貼合位置：玻璃載體或矽/玻璃中介層的最底層背面，並可能隨著RDL處理層數增加而多次使用。當 Silicon Interposer 經過背面研磨薄化至 50-100 微米時，結構已無法自我支撐，必須暫時鍵合在玻璃載具（Glass Carrier）上，請參考下圖。此時，Balance Film 就會改貼在「玻璃載具的最下方」，防止巨型載板在後續製程中因上下力矩失衡而碎裂。

作用機制：在載板正面不斷塗佈 PI/PBO 介電層與電鍍厚銅線路時，這些材料在高溫固化與冷卻後會產生強大的收縮力，導致載板向上捲曲（Smile 變形）或向下凸起（Cry 變形）。貼在背面的 Balance Film 會透過自身特製的高分子材料特性，在同溫層產生「反向的拉扯應力」，將載板強行拉平，確保後續黃光微影製程的精準對位。



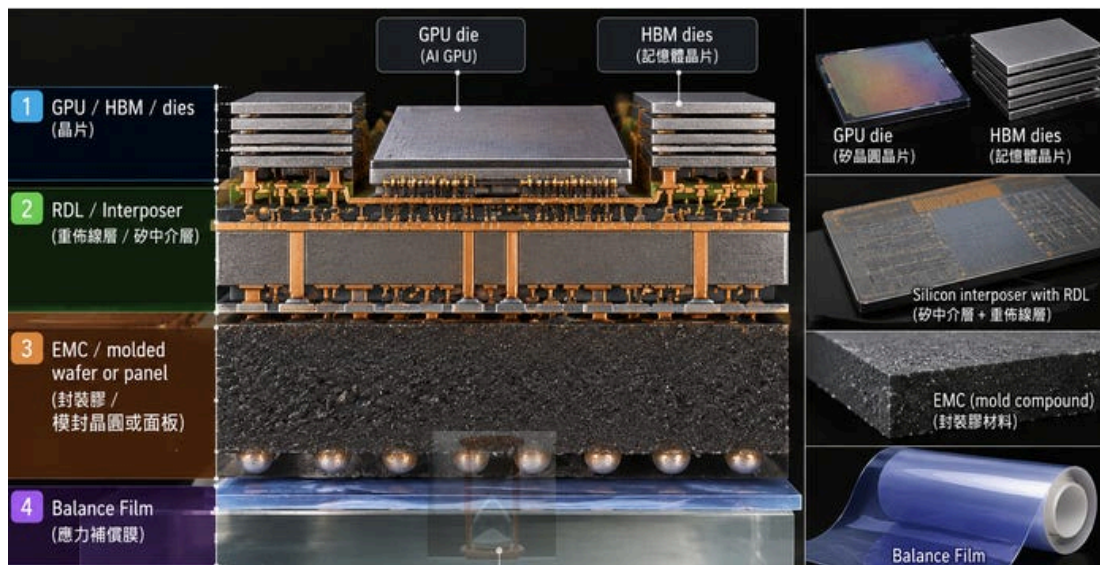
Source: TrendForce

2. Bump & Reflow（凸塊與迴錫）製程階段

貼合位置：環氧樹脂（EMC）載體的背面。

作用機制：當進行 Chip-first 或 Chip-last 製程進入高溫迴錫爐時，EMC 與晶片的膨脹率截然不同。此時 Balance Film 作為暫時性的應力平衡層貼合於背面，可抵消高溫帶來的劇烈形變，防止晶片移位或焊點斷裂。

總結來說：Balance Film 就像是拔河比賽中的另一方。正面（晶片與線路端）有多大的力量拉扯導致基板變形，背面的 Balance Film 就必須透過精準計算的材料特性（如模數、CTE 等），施加同等的力量將其拉回水平。它是一種具有功能性的暫時犧牲層，通常在製程完成、應力釋放或結構固定後，會透過解膠（De-taping）程序移除。



Source: Andrew Lu on global semis/techs

Avoiding warpage to return

Balance Film 的主要目的不是「永久消除應力」，而是搶救製程良率。微影機台的曝光對焦景深（DOF）通常只有數個微米的容忍度，只要基板翹曲超過極限，光阻就會失焦、走線就會斷裂。Balance Film 在關鍵幾道高溫化學品與電鍍工站維持住絕對水平，其任務就算圓滿完成。

雖然從產業鏈了解，目前台積電交給專做 CoS chip on substrate 封測廠的 CoW AI GPU/TPU 在出廠時即使不使用 Balance film 仍保持平整（部分原因是矽中介層還沒超過9個光罩尺寸），但時間一拉長，如果不貼上 Balance film，也沒有其他結構接手，殘留應力必然會導致翹曲「彈性回跳」

（Springback）。所以台積電希望 CoS 封測廠能共同分攤AI大晶片 Balance film 隨產品出貨的成本。但在實際先進封裝產線中，翹曲之所以不會毀掉成品，是因為隨產品出貨的 Balance Film 的核心使命是「階段性保駕護航」——它確保基板在最脆弱的微影與線路成形階段維持平整，一旦進入後段，應力的承載就會透過以下機制順利交棒：

1. 結構支撐接力（Structural Handoff）

- Balance Film 不會在基板毫無支撐的裸露狀態下被撕除。
- 通常是在晶圓/面板完成正面多層 RDL 製作後，先將正面與剛性載具（如第二層暫時載板、Bumping 載具）或次世代目標基板（如 ABF 載板）完成貼合或鍵合（Bonding），隨後才將背面的 Balance Film 解膠剝離。這時，結構的剛性已經由下游的承載體接手。

2. 加強框與散熱蓋物理鎖定（Stiffener Ring & Lid）

- 對於尺寸超過 80mm × 80mm 以上的大型 AI 晶片，封裝完成移除製程保護層後，本體仍有強烈的翹曲趨勢。

- 後段製程會立刻在封裝邊緣環氧樹脂表面貼上金屬加強框（Stiffener Ring），或是封上均熱片/散熱蓋（Lid），利用金屬件極高的機械剛性，直接將整個封裝體強制「鎖平」，避免在後續 SMT 焊接到主機板時發生焊點虛焊或開裂。

3. 對稱性結構補償（Symmetric Stack-up Design）

- Balance Film 解決的是「製程中單側堆疊引發的暫態應力失衡」。
- 在產品的終態設計中，工程師會在基板的上下兩側設計對稱的介電層（Dielectric Layer）與等效銅箔走線密度（Copper Ratio Balancing）。當封裝整體完成、所有高分子材料完全熟化（Cure）後，本體正反面的力矩會互相抵消，使結構達到自發性的力學平衡（Self-balancing）。

Story to be continued

在透過更多產業調研之後，我們之後打算出 Part II，來聊聊

- 有沒有更優的防翹曲技術？
- 產業鏈核心競爭力比較？
- 如何測算 Balance film 的市場規模及對營收貢獻的影響？

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Subject: AMD acquires World Labs to expand its AI Hardware-Software Co-Design
Date: 9 hours ago

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AMD acquires World Labs to expand its AI Hardware-Software Co-Design

AMD併購World Labs 的綜效及風險- 尤其加強軟硬兼施的AI晶片/系統設計及稀釋/商譽減損的風險

ANDREW LU ON GLOBAL SEMI/TECHS

SEP 28 • PAID



What's new?

AMD 以約 82 億美元全股票交易（All-Stock Transaction）收購由AI 教母李飛飛聯合創立的空間智能初創公司 World Labs，是其繼收購賽靈思（Xilinx, 500億美元）、Silo AI（6.65億美元）及 ZT Systems（49億美元）後，在 AI 領域另一前沿戰略布局。

依據協議，李飛飛將出任 AMD 執行副總裁兼首席科學家（EVP & Chief Scientist），直接向董事長兼執行長蘇姿丰（Lisa Su）匯報，核心團隊（包括共同創辦人 Justin Johnson 與 Ben Mildenhall）將組成 AMD 旗下的前沿 AI 研究機構。這項交易象徵 AMD 從純硬體晶片商向「全端 AI 平台（Full-Stack AI Platform）」的戰略轉移。



Synergies

1. Hardware-Software Co-Design)

過去 AMD 在 AI 戰場的最大痛點是「先定義晶片硬體，再由軟體團隊艱難適配 ROCm」，導致生態進度長年落後 NVIDIA CUDA。

- **模型導向架構 (Model-driven Silicon)**：World Labs 專注於空間智能 (Spatial Intelligence) 與大型世界模型 (Large World Models, LWMs)，涉及高並發的 3D 體積重建 (NeRF、3D Gaussian Splatting)、物理碰撞模擬及時空連續性計算。

付費訂閱戶可分享我們觀察到的3個併購綜效，4個風險，尤其是股權稀釋及商譽攤銷對股東的影響，尤其是未來的觀察指標

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- **反哺硬體規格**：頂級模型團隊直接進駐，能讓 AMD 在規劃下一代 Instinct 架構（如 MI400 系列及後續 UDNA 架構）時，精準針對底層張量核心、記憶體子系統（HBM 頻寬配置與容量）、晶片間互連（Infinity Fabric）進行客製化最佳化，擺脫被動追趕的窘境。

[晶片層 (Silicon)]	EPYC CPU + Instinct GPU (MI350/MI400) + DPU
↓	
[系統層 (Systems)]	ZT Systems (伺服器與超大規模叢集設計)
↓	
[生態層 (Software)]	ROCm 開源工具鏈 + Silo AI (企業客製落地)
↓	
[模型層 (Frontier AI)]	World Labs (李飛飛團隊：空間智能與世界模型)

2. Shifting focus to physical AI & embodied AI

當前以文字與 2D 圖像為核心的生成式 AI 正逐步成熟，下一階段的兵家必爭之地是理解物理規律、三維環境模擬與機器人交互。

- **抗衡 NVIDIA Omniverse 與 Cosmos**：NVIDIA 憑藉 Omniverse 數位孿生與 Cosmos 世界基礎模型，深度綁定自駕車、機器人與工業製造業者。AMD 過去在此領域近乎空白，World Labs 的加入補齊了關鍵資產，使 AMD 具備建構開放式物理世界模擬平台的基石。

3. Buying elite AI software engineers

- **人才磁吸效應**：李飛飛在人工智慧與計算機視覺領域具備學術威望。由其擔任首席科學家，能大幅扭轉 AMD「工程實力強但缺乏 AI 頂級科學家陣容」的市場認知，對後續吸納全球頂尖演算法研究員與跨界人才具備指標性意義。
- **推動開源社群擴張**：World Labs 團隊的入駐，能從演算法源頭優化 ROCm 開源軟體堆疊對前沿多模態架構的支援，降低開發者在 AMD 硬體上訓練與微調複雜模型的阻力。

Risks and challenges

維度	主要風險因子	實質影響評估
財務估值	起高溢價收購與股權稀釋	82 億美元全股票交易約稀釋現有股權 3%–4%；World Labs 成立僅約兩年，存在龐大商譽（Goodwill）減損隱憂。
商業化落地	空間智能變現週期漫長	3D 世界模型尚處前沿探索階段，短期內無法貢獻實質營收，高昂算力研發開支恐壓抑營業利率。
組織文化	半導體工程 vs. 學術研究機構	嚴謹的 Tape-out 交付節奏與前沿開放研究節奏存在摩擦，核心科學家在鎖定期後的留任意願存疑。
客戶關係	與下游 CSP/模型客戶潛在競合	由「中立算力供應商」跨足「自研基礎模型」，可能引發微軟、Meta 等雲端大客戶的邊界疑慮。

Source: Andrew Lu on global semis/techs

1. 財務溢價過高與商譽減損風險

World Labs 於 2024 年成立，早期融資估值約在 10 億美元左右，本次 AMD 支付 82 億美元，隱含極高的收購溢價。

- 儘管採全股票收購避免了現金儲備消耗，但會直接稀釋流通股東權益。
- 若未來 3 至 5 年內該技術未能實質帶動 Instinct 晶片或系統出貨量激增，AMD 將面臨商譽（Goodwill）與無形資產減損的提列壓力。我們目前預測 AMD 未來5年每年將額外攤銷15億美元軟體資產溢價，讓年度折

舊及攤銷從30億美元增加50%到45億美元，除了3-4%股權稀釋外，影響0.9美元年度GAAP EPS 或2027年度彭博社分析師預期的 13.66美元 GAAP EPS 的6.6%，但未來5年 Non-GAAP EPS 預測不受影響。

2. 研發燃燒率（R&D Cash Burn）與商業化空窗期

世界模型與物理模擬訓練需要極端龐大的算力資源與工程投入。空間智能目前距離廣泛的企業級商業變現仍有一定距離，短期內 World Labs 不僅無法貢獻自由現金流，還需佔用大量內部 GPU 叢集與研發預算，可能在短期內對 AMD 的非 GAAP 營運利潤率（Operating Margin）造成壓力。

3. 組織文化整合與頂尖人才留任

- 硬體半導體企業的文化核心是「嚴格的產品進度表、良率提升與供應鏈管控」；而前沿 AI 實驗室的核心則是「開放式論文發表、自由度極高的算力探索」。
- 過去大廠收購純研究型 AI 團隊後，常因官僚流程或研究自由受限導致核心人才流失。李飛飛團隊能否在體量龐大的半導體公司架構內保持研究活力，取決於管理層授予的獨立性與激勵機制設計。

4. 與下游客戶的競合衝突（Channel Conflict）

- AMD 先前定位是純粹的中立算力供應商，力求承接微軟、Meta、AWS 等不想被 NVIDIA 壟斷的轉單需求。
- 當 AMD 自身具備由李飛飛領軍的前沿模型團隊後，如何在「自研前沿模型技術」與「支援客戶自研模型」之間維持中立界線，需謹慎拿捏，避免讓雲端巨頭與下游 AI 新創產生戒心。

The points in future to check

後續評估此項併購成效的核心觀察點包含：

1. **硬體轉化速度**：World Labs 的演算法研究能否在 12-18 個月內實質轉化為 AMD UDNA/ROCm 上的專屬優化庫，並直接體現在 MI400 等下一代晶片的基準測試（Benchmarks）上。
2. **人才留存率**：李飛飛及其核心研究團隊在交易完成後的留任年限與論文/架構產出穩定度。
3. **生態聯盟建立**：AMD 是否能以 World Labs 為核心，聯合自駕車、機器人、遊戲引擎（如 Epic/Unity）及工業製造巨頭，建立抗衡 NVIDIA Omniverse 的開源標準體系。

這項交易本質上是 AMD 在完成晶片與系統（ZT Systems）拼圖後，向 AI 價值鏈最高端「認知架構與世界模型」邁出的防禦兼進攻型豪賭；若整合順暢，AMD 將具備從底層矽智財到實體 AI 基礎模型的端到端定義能力，但其昂貴的代價與漫長的技術轉化週期，將高度考驗管理層的整合與資本配置能力。

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